Indian Institute of Technology Ropar
Department of Mathematics
MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 4

Solve the following LPP using the Big-M method

1. **Maximize** $Z = 3x_1 + 2x_2$

   Subject to constraints

$$2x_1 + x_2 \leq 12,$$
$$3x_1 + 4x_2 \geq 12,$$
$$x_1, x_2 \geq 0.$$

2. **Maximize** $Z = 3x_1 + 2x_2 + x_3$

   Subject to constraints

$$2x_1 + 5x_2 + x_3 = 12,$$
$$3x_1 + 4x_2 = 11,$$
$$x_2, x_3 \geq 0, \quad x_1 \text{ unrestricted in sign.}$$

3. **Maximize** $Z = 6x_1 + 4x_2$

   Subject to constraints

$$2x_1 + 3x_2 \leq 30,$$
$$3x_1 + 2x_2 \leq 24,$$
$$x_1 + x_2 \geq 3,$$
$$x_1, x_2 \geq 0.$$

4. **Maximize** $Z = 3x_1 + 5x_2$

   Subject to constraints

$$x_1 - 2x_2 \leq 6,$$
$$x_1 \leq 10,$$
$$x_2 \geq 1,$$
$$x_1, x_2 \geq 0.$$

5. **Minimize** $Z = 5x_1 + 3x_2$

   Subject to constraints

$$2x_1 + 4x_2 \leq 12,$$
$$2x_1 + 2x_2 = 10,$$
$$5x_1 + 2x_2 \geq 10,$$
$$x_1, x_2 \geq 0.$$